

hoax, partial truth, and overt lie) plus 7 control pairs for non-deceptive behaviors.

2. **Framing Variations:** We generated 34 additional prompt pairs using Claude Sonnet 4.5 (Anthropic, 2025), exploring different ways to linguistically frame the honest/dishonest distinction, without targeting a specific deception type. Categories include identity-based framings (e.g., “You are an AI designed to be fundamentally truthful”), role-based framings (e.g., “Your role is to provide factual and accurate information”), instruction-based, hybrid, control repetition, long-form, and baseline framings. These explore whether the surface-level structure of the prompt affects probe performance independently of the deception type being targeted.

3.5. Baselines and Skylines

For baselines, we use an instruction-pair probe with the default system prompt from Goldowsky-Dill et al. (2025) (described in Section 3.1). We also computed in-distribution skyline probes on each evaluation dataset using an 80/20 train/test split, representing the best achievable performance when the probe’s training distribution matches the evaluation distribution. These skyline probes achieved near-perfect AUC (≥ 0.993 on nine of ten datasets), and are omitted from the main figures for clarity.

3.6. Training of Probes

We train logistic regression probes on the residual stream values extracted at layer 20 (zero-indexed) of the model (as described in Section 3.1). These activation vectors are standardized to zero mean and unit variance across the dataset before training. We aggregate over all response tokens except the final 5, using the mean. The L2 regularization constant was tuned as part of the ANOVA sweep; we found $\lambda = 1$ to be optimal.

4. Experiments

4.1. Analysis of Variance across Training Setups

We identified four factors that affect probe performance: (1) the instruction pair (honest/deceptive prompt) used when training the probe, (2) the specific dataset of true facts used as the model response, (3) the token selection strategy for aggregation from the model response (first 5, first 10, all except last 10, all except last 5, or chat template tokens only), and (4) the layer probed at.

In order to determine where to focus our efforts, we trained 2000 probes using different combinations of these factors (exploring layers 20, 25, and 31), and evaluated each on the validation dataset described in Section 3.3. We then

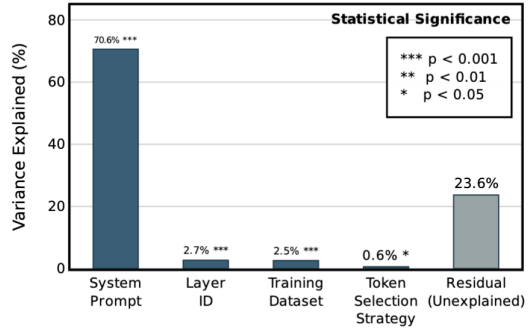


Figure 1. Variance in AUC on evaluation datasets explained by experimental factors. System prompt explains 70.6% of performance variance ($p < 0.001$), substantially exceeding variance explained by layer ID (2.7%), training dataset (2.5%), or token selection strategy (0.6%). Residual unexplained variance accounts for 23.6%.

conducted an analysis of variance (ANOVA) on the results to determine the fraction of variance attributable to each factor.

Most of the variance in probe performance is attributable to changes in the system prompt, with relatively minor effects caused by variation in the layer ID, training dataset, or token selection policy. There are minor interaction effects between the system prompt and training dataset, but in general they are dominated by the system prompt variations.

This result forms our justification for focusing on system prompt variation and fixing the remaining factors for the remainder of our analysis. Given that layer choice explained only 2.7% of variance, we selected layer 20 for all subsequent experiments. We also fixed: 1) Token selection to all tokens up to 5 tokens before the end of the response. 2) Training dataset to the original dataset of true and false facts from Azaria & Mitchell (2023)

Cross-model replication on Llama 3.3 70B Instruct. To assess whether the dominance of system prompt choice generalizes beyond Gemma-2-9B-IT, we replicated the same ANOVA sweep on Llama-3.3-70B-Instruct. System prompt choice again dominates, explaining 71.0% of performance variance ($p < 0.001$), compared with 5.8% for layer ID, 4.5% for training dataset, and 0.3% for token selection strategy (Figure 2).

4.2. Probe Training with Varying System Prompts

The ANOVA result motivates a key question: given that prompt choice dominates probe performance, do different deception datasets require different prompts, or does a sin-

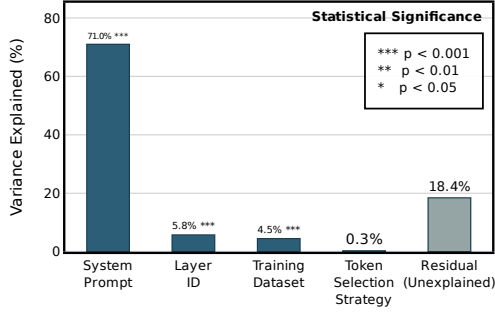


Figure 2. ANOVA variance decomposition on Llama-3.3-70B-Instruct. System prompt choice explains 71.0% of performance variance, consistent with the 70.6% observed on Gemma-2-9B-IT (Figure 1).

gle well-chosen prompt suffice? We hypothesize that deception is sufficiently heterogeneous that specialized probes will outperform a single generic detector. To test this, we train separate probes for each system prompt variation and evaluate them across all datasets, following the validation procedure described in Section 3.3.

5. Results

Our experiments confirmed that the system prompt provided to the model is the dominant factor affecting deception probe performance. We observed substantial performance variation across probes trained with different system prompts, when training probes with the system prompt, training dataset, token selection and layer ID varied independently. Importantly, the best-performing probes were interpretable, with clear semantic relationships between the prompt content and the types of deception they successfully detected. We identified specific system prompts that achieved optimal performance for different datasets, reflecting the underlying heterogeneity in deception types. This finding suggests that detecting deception across datasets requires specialized detection approaches rather than a one-size-fits-all solution.

5.1. Comparison of Performance Across Probe Types

We evaluated the effectiveness of deception taxonomy-based prompts (Druckman & Bjork, 1991) for deception detection by comparing these prompts to other probe variants across ten deception datasets. Using linear probes trained on Gemma-2-9B-IT layer 20 representations, we assessed:

1. *Deception Baseline* prompts using generic hon-

est/dishonest distinctions;

2. the single *Best Average* probe (best mean performance across all datasets without using taxonomy);
3. the *Best Taxonomy* probe for the specific dataset being evaluated.

All ten datasets had improved performance with a specific taxonomy prompt: eight exceeded a 5% AUC gain, three exceeded 10%, and one (MASK known facts) exceeded 25%. Importantly, for every dataset there existed a taxonomy-based probe that outperformed the baseline.

Dataset-specific performance patterns. The magnitude of improvement from taxonomy-based prompts varies continuously across datasets, ranging from near-zero to over 86% relative gains. This variation reflects both the baseline detectability of different deception types and the degree to which specific taxonomy prompts match the underlying deception mechanism.

At one extreme, *liars-bench-convincing-game* shows minimal improvement (Baseline: 0.984, Taxonomy: 0.985, $\Delta = +0.001$), suggesting that straightforward lies where the model is explicitly prompted to lie exhibit stereotypical patterns easily detected even with generic prompts. At the other extreme, *mask-known-facts* demonstrates dramatic gains ($0.374 \rightarrow 0.697$, $\Delta = +0.323$, +86.3%) when using pressure-dishonesty prompts (AUC = 0.697), indicating that deception under social pressure requires specifically targeted detection approaches.

Several datasets show substantial improvements exceeding 20% relative gains: *mask-doubling-down-known-facts* ($0.565 \rightarrow 0.727$, +28.7%) and *sandbagging* ($0.518 \rightarrow 0.657$, +26.8%), which responds particularly well to concealment-focused prompts. Meanwhile, datasets like *mask-disinformation* (Baseline: 0.842, Taxonomy: 0.901, $\Delta = +0.059$) show more modest but meaningful improvements when using prompts targeting information fabrication. The remaining datasets show consistent gains in the 6-13% range.

These patterns suggest that detection difficulty depends on both deception complexity and the availability of well-matched taxonomy prompts. We hypothesize that datasets involving nuanced, context-dependent deception benefit most from targeted prompts, while more explicit deception types are easier to detect generically. This heterogeneity suggests that ensemble methods combining multiple specialized probes may be more effective than seeking a single universal detector. Deployment environments may benefit from determining the most important threat model to focus on when deploying probes.

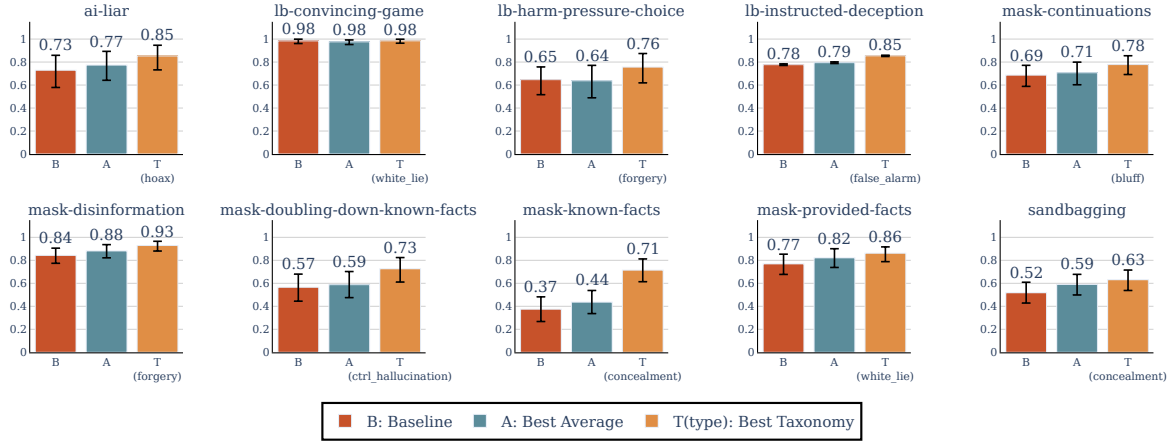


Figure 3. Comparative AUC across all datasets and approaches. Results show a consistent hierarchy: Deception Baseline < Best Average < Best Taxonomy. The Baseline achieved mean AUC = 0.689 (SD = 0.165). A single Best Average probe selected on a validation set achieves mean AUC = 0.721 ($\Delta = +0.032$, +5.5%), demonstrating limited universal detection capability. Post-hoc oracle analysis (Best Taxonomy) reveals a performance ceiling of mean AUC = 0.797 ($\Delta = +0.108$, +20.5%) when probes are perfectly matched to deception types. The gap between Best Average and Best Taxonomy highlights the importance of matching probes to specific threat models; synthetic validation experiments (Section 5.2) confirm this ceiling is achievable *a priori*.

5.2. Predictive Validation of Taxonomy-Targeted Probes

The post-hoc oracle results raise a key question: can we achieve strong detection performance when we specify the deception type *a priori*? We conducted two analyses to provide predictive evidence that taxonomy-based probes preferentially target their intended deception type and that probe rankings transfer to held-out data.

Synthetic datasets with known deception type. We constructed synthetic LLM-generated evaluation datasets for each of the 16 deception types. For each scenario, we prompted the model to engage in (or not to engage in) the specific taxonomy behavior, with wording distinct from the training system prompts, yielding an even split of honest and deceptive responses for that taxonomy type. Testing all probes against all synthetic datasets, the matched probe ranks in the top 3 for its own deception type in 8 of 16 cases, with near-perfect AUCs for several (*evasion*:0.999, *hoax*:0.993, *forgery*:0.989). The matched probe also beats the baseline probe in 12 of 16 cases. We note that in particular for the *counterfeit* category, which showed a distinct direction in the clustering analysis (Section 5.3), the targeted probe greatly outperformed the baseline. The full probe-by-synthetic-dataset AUC heatmap is shown in Figure 4.

Split-half stability. For each evaluation dataset, we split the data 50/50, select the best taxonomy probe on one half, and evaluate it on the held-out half, repeating across 10 random seeds. The probe selected on the first split

ranks #1 on the second split in 45% of cases (chance = 6.25% with 16 probes) and ranks in the top 3 in 67% of cases. The mean Spearman rank-order correlation of probe orderings across splits is $\rho = 0.69$. Stability is strongest on datasets where probes perform well overall (e.g., *liars-bench-instructed-deception*: same probe selected 10/10 seeds, rank 1 on the held-out split every time, $\rho = 0.99$; *mask-known-facts*: same probe 9/10 seeds, $\rho = 0.74$) and weakest where absolute AUCs are low (e.g., *mask-doubling-down-known-facts*: $\rho = 0.20$).

5.3. Clustering Probe Scores to Identify Shared Mechanisms

To understand whether different taxonomy prompts detect distinct deceptive mechanisms or converge on shared underlying patterns, we analyzed the pairwise correlations between probe outputs across all evaluation samples. We computed Pearson correlations for all probe pairs and applied hierarchical clustering to this correlation structure, identifying five distinct clusters of probes with similar detection patterns (Table 1). The full correlation matrix is shown in Figure 7 in the Appendix.

The relatively high correlations within clusters reveal that despite using different linguistic framings, many taxonomy prompts detect the same underlying mechanisms in the model’s activation space. For instance, Cluster 1 probes (*evasion*, *exaggeration*, *false_alarm*, *forgery*, *trick*,